

Tutorial - Week 7

TG 11 - TG 12

1. Define with a single formula a function f such that:

- The domain of f is $[0, 3) \cup (3, 5) \cup (5, +\infty)$,
- f has exactly one vertical asymptote and one horizontal asymptote,
- f has exactly one local maximum which is also global, and one local minimum but no global minimum.

Hint: sketch the graph first.

2. Sketch the graph of f .

- (a) $f(x) = \frac{x^2-4}{x-1}$.
- (b) $f(x) = \frac{x^2-2x+1}{x^2-2x-3} + 2$.
- (c) $f(x) = 1 - x + e^{1+\frac{x}{3}}$.
- (d) $f(x) = \arctan\left(\frac{x-1}{x+1}\right)$.
- (e) $f(x) = \begin{cases} (1 - \cos x)^{\frac{1}{3}} & \text{if } -\frac{7\pi}{2} < x < 0, \\ \arctan(x^2 - 3x + 2) & \text{if } x \geq 0. \end{cases}$

Beautiful Dance Moves



3. Which point of the line segment $C = \{(x, y) \in \mathbb{R}^2 : x + y = 1, -1 \leq x \leq 0\}$ is closest to the point $(1, \sqrt{3})$?

4. Find an equation of the line through the point $(3, 5)$ that cuts off the least area from the first quadrant.

5. Find the general antiderivative of the function.

- (a) $f(x) = x^{\frac{1}{3}} + (3x)^{\frac{1}{3}}$.
- (b) $f(x) = 2 \sin(x) + \cos(3x)$.
- (c) $f(x) = \frac{3x}{1+9x^2}$.
- (d) $f(x) = \frac{1}{2}e^{-4x} + \cosh x$.
- (e) $f(x) = 8 \sec x (\sec x - 4 \tan x)$.
- (f) $f(x) = \frac{1}{\tan x}$.

6. Solve the initial value¹ problem.

- (a) $f'(x) = x^{-4}, f(1) = 1$.
- (b) $f'(x) = 3x^5 - 6x^2 + 15x - 1, f(0) = -1$.
- (c) $f'(x) = -\cos x + \sec^2 x, f\left(\frac{\pi}{4}\right) = 2 + \frac{\sqrt{2}}{2}$.
- (d) $f''(x) = x^3 + \sinh x, f(0) = 3, f(2) = 2.6$.

7. Estimate the area under the graph of $f(x)$ from $x = 0$ to $x = 1$ using the following:

¹Student: "But the only initial value I have is my stress level: 100%" 😊

(a) R_{20} .(c) M_{20} .(e) L_{100} .(b) L_{20} .(d) R_{100} .(f) M_{100} .

8. Find the following definite integrals using Riemann sums:

(a) $\int_1^2 x^3 dx.$

(c) $\int_{-1}^2 x^3 - x^2 dx.$

(b) $\int_0^1 e^x dx.$

9. Given $\int_{-2}^3 f(x) = 2$, $\int_{-2}^0 g(x) = -1$ and $\int_3^0 f(x) = -\frac{2}{3}$, compute the following integrals.

(a) $\int_{-2}^0 f(x) dx.$

(b) $\int_{-2}^0 2f(x) + \frac{1}{3}g(x) dx.$

10. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Show that f is not integrable.

